

# Model review and revisions

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I reviewed the model and calibration, including preferences, credit, demographic accounting, and the empirical targets.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Tested equivalence scales and changing housing shares.
- Rebuilt the first-birth housing evidence and national input measures.
- Revised adult entry and tested how purchase timing affects the calibration.

# Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

## Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[ \frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left( \frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here  $h$  is rooms,  $m$  is children at home, and  $\tau \in \{R, O\}$  denotes tenure. Parents require  $h > h_P$ ;  $\chi$  raises housing services for owners. The equivalence scale  $e(m)$  increases family consumption needs. Child benefits are zero at  $m = 0$ ;  $\psi$  sets their level and  $\gamma$  controls diminishing marginal benefits.

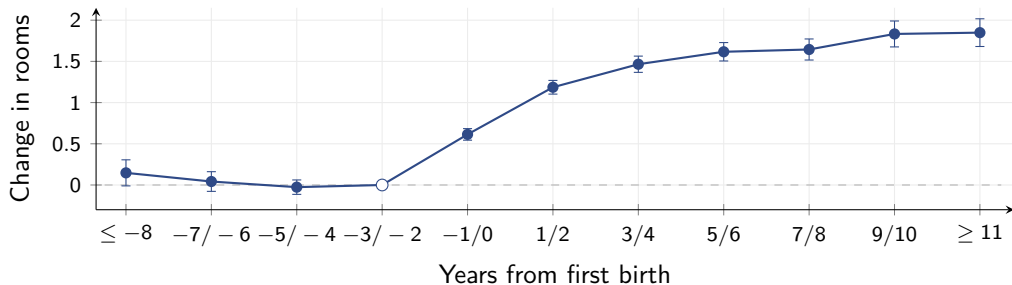
# Demographic and market equilibrium

Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines  $H_0$ .
- A balanced payroll tax finances equal retiree pensions.

## Housing around the first birth

We found a mistake in the earlier regression and rebuilt the PSID event study using interview dates and household status before birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years  $-3/-2$ ; person and year effects; survey weights; 95% pointwise intervals.

## Credit timing

A renter with financial wealth  $b$  and period income  $y$  who buys a home worth  $Q = pH$ :

$$c + (\delta + \tau_H) Q + b' = R(b - Q) + y.$$

Normally the buyer must have the 20% down payment at closing:  $b \geq (1 - \phi)Q$ , with  $\phi = 0.8$ .

**The problem.** A model period is four years. Requiring the down payment from wealth at the start of the period ignores saving within the period; counting all of it lets a buyer close with little equity and reach 20% only over the next four years. The rules differ in how much of the period's net saving  $S = b' - (b - Q)$  counts:

$$b + \lambda S \geq (1 - \phi) Q.$$

- Soft,  $\lambda = 1$ : all of the period's saving counts.
- Quarter-saving,  $\lambda = \frac{1}{4}$ : a quarter of it counts, roughly one year of saving.
- Hard,  $\lambda = 0$ : only wealth brought into the period counts.

## Externally set parameters

| Parameter                            | Value                | Source                         |
|--------------------------------------|----------------------|--------------------------------|
| Model period                         | 4 years              |                                |
| Entry and retirement ages            | 18, 66               |                                |
| Risk aversion $\sigma$               | 2                    |                                |
| Equivalence scale $(\omega, \nu)$    | (0.7, 0.7)           | Scholz et al. (2006)           |
| Consumption share $\alpha_0$         | 0.733                | CEX 2019–23, childless renters |
| Annual real interest rate            | 2%                   | Greaney et al. (2025)          |
| Annual depreciation; property tax    | 1.416%; 1.060%       | National housing inputs        |
| Financed share $\phi$                | 0.80                 | 20% down payment               |
| Selling cost $\psi_s$                | 6%                   | Greaney et al. (2025)          |
| Owner house sizes                    | 2, 4, 6, 8, 10 rooms |                                |
| Largest rental unit                  | 6 rooms              | Greaney et al. (2025)          |
| Housing-supply elasticity            | 0.63                 | Baum-Snow and Han (2024)       |
| Pension / working-household earnings | 0.229                | CPS ASEC, income 2007          |
| Payroll tax                          | 8.0%                 | Balances the pension budget    |
| Bequest wealth shift $\theta_1$      | 0.008                | 1% of median annual earnings   |

# Parameters and target moments

2007 stationary economy: ten parameters, eleven moments.

| Parameter                  | Interpretation               | Main target moment                        | Target |
|----------------------------|------------------------------|-------------------------------------------|--------|
| <i>Saving and bequests</i> |                              |                                           |        |
| $\beta$                    | Patience                     | Wealth / earnings                         | 6.927  |
| $\theta_0$                 | Bequest weight               | Annual bequests / wealth (%) <sup>*</sup> | 0.729  |
| <i>Housing and tenure</i>  |                              |                                           |        |
| $\chi$                     | Owner housing premium        | Ownership, ages 30–55 (%)                 | 67.626 |
| $\kappa_T$                 | Tenure tastes                | Recent-parent ownership difference (pp)   | 12.761 |
| $h_P$                      | Basic space for parents      | First-birth rooms difference <sup>*</sup> | 1.465  |
| –                          | Additional moment            | Mean rooms <sup>*</sup>                   | 5.729  |
| <i>Fertility</i>           |                              |                                           |        |
| $\psi$                     | Child-benefit level          | Completed fertility                       | 2.100  |
| $\xi$                      | First-birth cost             | Childlessness, ages 40–44 (%)             | 19.828 |
| $\kappa_1$                 | First-birth taste dispersion | Mean first-birth age                      | 25.976 |
| $\gamma$                   | Child-benefit curvature      | One child among mothers (%)               | 21.366 |
| $\kappa_C$                 | Later-birth taste dispersion | Children at age 25                        | 0.810  |

Each parameter is paired with the moment it mainly moves; all ten are estimated jointly. <sup>\*</sup>Recomputed: mean rooms (AHS), bequests (SCF), first-birth rooms (PSID).



# Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

| Moment                                      | Target       | Soft         | Hard         | Quarter-saving |
|---------------------------------------------|--------------|--------------|--------------|----------------|
| Completed fertility                         | 2.100        | 2.100        | 2.100        | 2.100          |
| Childlessness, ages 40–44 (%)               | 19.828       | 20.222       | 20.413       | 20.546         |
| One child among mothers, ages 40–44 (%)     | 21.366       | 21.264       | 20.907       | 20.844         |
| Mean first-birth age (years)                | 25.976       | 26.032       | 26.007       | 26.057         |
| Wealth / earnings                           | 6.927        | 6.551        | 6.610        | 6.645          |
| Annual child bequests / wealth (%)          | 0.729        | 0.685        | 0.684        | 0.685          |
| <b>Mean occupied rooms</b>                  | <b>5.729</b> | <b>5.957</b> | <b>6.241</b> | <b>6.131</b>   |
| Ownership, ages 30–55 (%)                   | 67.626       | 66.527       | 60.878       | 64.572         |
| <b>First-birth rooms difference</b>         | <b>1.465</b> | <b>1.324</b> | <b>1.063</b> | <b>1.167</b>   |
| Recent-parent ownership difference (pp)     | 12.761       | 11.825       | 10.125       | 11.683         |
| <b>Children at age 25 (capped at three)</b> | <b>0.810</b> | <b>0.528</b> | <b>0.524</b> | <b>0.521</b>   |
| Validation moments                          | Target       | Soft         | Hard         | Quarter-saving |
| First births at age 30+ (%)                 | 24.928       | 23.448       | 23.078       | 23.395         |
| Older wealth/income p90/p50                 | 3.516        | 2.968        | 2.861        | 2.937          |
| Family rooms difference                     | 0.385        | 0.304        | 0.321        | 0.316          |

## Parameter estimates

Estimates under the three purchase rules;  $H_0$  follows from housing clearing.

| Parameter  | Economic interpretation                     | Soft  | Hard  | Quarter-saving |
|------------|---------------------------------------------|-------|-------|----------------|
| $\beta$    | Annual discount factor                      | 0.967 | 0.967 | 0.968          |
| $\chi$     | Owner housing-service premium               | 1.097 | 1.068 | 1.091          |
| $\xi$      | First-birth fixed utility cost              | 0.352 | 0.385 | 0.397          |
| $\kappa_1$ | First-birth taste dispersion                | 0.124 | 0.123 | 0.128          |
| $\kappa_C$ | Later-birth taste dispersion                | 0.363 | 0.350 | 0.348          |
| $\theta_0$ | Bequest weight                              | 0.106 | 0.106 | 0.109          |
| $\gamma$   | Child-benefit curvature                     | 0.101 | 0.101 | 0.098          |
| $\kappa_T$ | Tenure-choice dispersion                    | 0.014 | 0.014 | 0.015          |
| $\psi$     | Child-benefit level                         | 0.178 | 0.172 | 0.174          |
| $h_P$      | Basic space requirement for parents (rooms) | 2.504 | 2.562 | 2.492          |
| $H_0$      | Housing supply coefficient                  | 6.757 | 7.546 | 7.289          |

## More credit raises ownership but not births

Financed share 80% vs. 100%, house price held fixed

|                                         | 80%  | 100% |
|-----------------------------------------|------|------|
| Ownership, ages 30–55 (%)               | 64.6 | 74.0 |
| Ownership, ages 18–21 (%)               | 10.5 | 31.4 |
| First-birth probability, ages 18–21 (%) | 25.5 | 25.8 |
| Completed fertility                     | 2.10 | 2.09 |

Households aged 18–21 have just entered the model, with the same wealth and income in both cases. For them the difference comes from financing alone.

Recalibrating under the hard and quarter-saving rules gives the same result: first births fall by about 0.5%.

## Why the response is small: what we have checked

- Rent per room equals the owner's cost per room, and  $\chi$  does not depend on children. Credit makes buying easier but does not make a child's space cheaper.
- 60% of renting parents are at the six-room cap. Without the cap the median would rent 7.3 rooms. Holding saving fixed, the cap costs them about 2% of consumption in that period.
- The parent requirement  $h_P$  is the same for one, two or three children.

These are candidate explanations. We have not yet shown how much of the small response each one accounts for.

## What I want to try next

- Selling costs: check whether the cost of trading up after a birth is what keeps credit from reaching fertility, and whether the 6% seller-paid cost is the right convention.
- Small owner homes: young buyers may lock into homes too small for a family.
- A housing requirement that grows with the number of children.
- Recalibrate before drawing conclusions from any of these.